
AR 516 : Introduction to Blockchain and Web3

Course Code : AR 516

Course Name : Introduction to Blockchain and Web3

L-T-P-C : 3-0-0-3

Intended for : Undergraduate (Elective)/ Postgraduate

Prerequisite : Knowledge of basic computer terminology

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Blockchain Technology Introduction (8 hours)**
 - What is Blockchain
 - How it evolved
 - Key Terminologies – Ledger, Smart Contract, Peer Network, Wallet, Hashing, Cryptography, Distributed vs Centralized Database, Blocks, Merkle Trees
 - Popular Blockchain Variants – Bitcoin, Ethereum, Hyperledger
 - Consensus Algorithms
 - Types of Blockchain
 - SideChains
- **Cryptocurrency and Bitcoin Introduction (6 hours)**
 - Transformation of Financial Systems
 - How current System Works and Related Problems
 - Distributed Ledger Technology
 - Bitcoin Evolution
 - Bitcoin Ecosystem
 - Bitcoin Clients and Wallets
 - Buy and Sell Bitcoin
 - Bitcoin Transaction and Structure
- **Course outline – Bitcoin Details and Mining Process (4 hours)**
 - Initiating Bitcoin Transaction
 - Transaction Propagation in Distributed Network
 - Role of Miners
 - Block Structure of Bitcoin
 - Merkle Tree Formation
 - Transactions Storage in Digital Ledger
 - Mining Algorithms – Proof of Work

- Consensus Algorithms
 - Fraud and Hack Inversion mechanism
 - Other Mining Algorithms – Proof of Stake, PBFT
 - **Overview of Ethereum Blockchain** (6 hours)
 - Introduction to Ethereum and differences from Bitcoin
 - Introduction to Smart contracts and solidity
 - Ethereum tools and development eco-system
 - Solidity language constructs and programming examples
 - Ethereum clients and Wallets
 - Layer 1 – Layer 2 differences and development
 - **Overview of Hyperledger Fabric Blockchain** (4 hours)
 - Enterprise Blockchain and Linux foundation
 - Architecture of Hyperledger Fabric
 - Network / explorer / installation.
 - Smart contract using Go / Node.JS
 - Other Enterprise Blockchains Course outline – Blockchain Applications and Hands on (10 hours)
 - Industry wide applications of Blockchain / web3
 - Current market size / outlook / industry trends
 - Career options and research opportunities
1. **Practical application 1** : Defi based application from finance sector
 2. **Practical application 2** : Smart contract / token based application for multiple sectors
 3. **Practical application 3**: Trade Bot / AI based prediction models for Crypto.
 4. **Practical application 4**: Blockchain based Gaming application.

Textbooks:

1. Julie EG, Nayahi JJ, Jhanjhi NZ, editors, **Blockchain Technology: Fundamentals, Applications, and Case Studies**, CRC Press.
2. Li KC, Chen X, Jiang H, Bertino E, editors, **Essentials of Blockchain Technology**, CRC Press.

References:

1. Banafa, A., **Blockchain Technology and Applications**, CRC Press, 2022.
2. Laurence, T., **Introduction to Blockchain Technology**, Van Haren, 2019.
3. Kim S, Deka GC, editors, **Advanced Applications of Blockchain Technology**, Springer, 2020.